

1114L

Management Studies Tripos

**ELECTIVE MODULE
COURSEWORK**

MSE12

Macroeconomics

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Introduction

Since the end of 2019, Covid-19 has rapidly spread into a global pandemic directly causing over 2.7m deaths (John Hopkins, 2021). The number of known cases is now over 120m (ibid), due to the transnational labour flows which characterise our global economy. The widespread economic effects of the virus have caused the first global recession since 2009 (World Bank, 2020).

This report will explain the mechanisms causing these effects, linking data and theory. It will then discuss the longer-term effects of Covid-19 on economic growth through physical, human, and social capital effects. Then, the determining factors which should shape policy responses to mitigate against long-term damage will be explored.

Short term: doom and gloom

The scale of the fall in global output has been astonishing, creating the deepest global recession since World War II (World Bank, 2020). The primary effects of the pandemic upon an economy are a reduced labour force due to death and illness, in addition to behavioural changes by agents. These effects have caused output to fall via negative shocks to both short run aggregate supply (SRAS) and aggregate demand (AD).

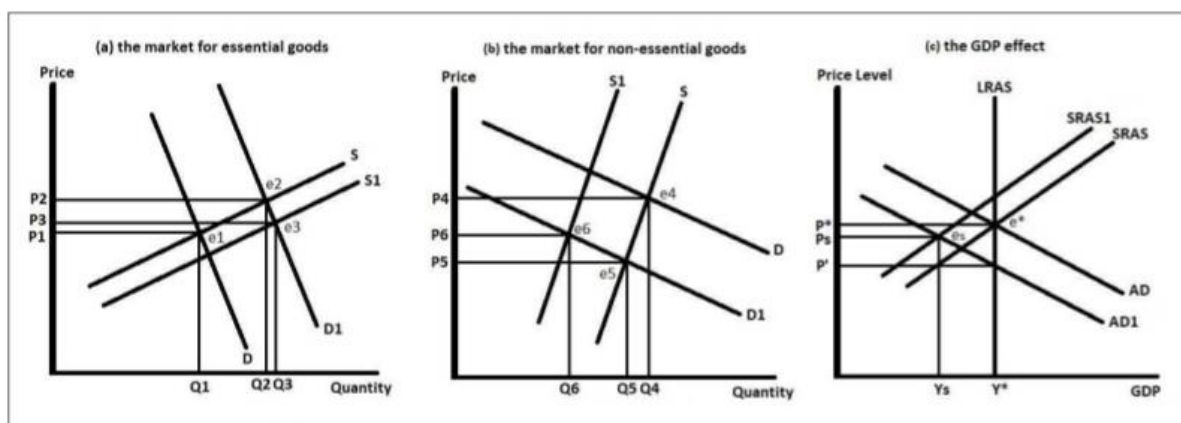
In the UK, there have been over 4.3m cases of Covid-19 (OWID, 2021). Infections reduce labour supply and productivity (Mohaddes et al., 2020) due to the virus' symptoms and self-isolation rules (some workers cannot perform their tasks remotely, and some do so less productively). This loss of labour inputs, coupled with government regulations, has forced many businesses to operate at reduced capacity, thus reducing SRAS. Obvious examples include fast-food restaurants offering reduced menus and airlines offering limited routes. Similarly, new laws have forced the closure of businesses in specific sectors, such as entertainment and hospitality, to slow the spread of the virus. Another factor causing SRAS to fall is the disruption of global supply chains due to increased border regulations. This has disrupted businesses relying on imported inputs, with 94% of Fortune 1000 companies have seen supply disruptions due to Covid-19.

Our focus now shifts to aggregate demand, which in the UK is composed mostly (66%) of consumption (World Bank, 2020). Therefore, UK consumption falling by 21% from Q1 to Q2, 2020 has contributed significantly to gross domestic product (GDP) falling by 24% across the same period (HoC, 2021). Most of this sharp fall in consumption is explained by lower incomes and animal spirits.

As mentioned, many firms have suspended operations during the pandemic. These firms still face fixed costs such as mortgages payments, but have a reduced revenue stream to cover these costs. To survive, many have had to cut variable costs by laying off staff, which has caused UK unemployment to rise from under 4% to over 5% during 2020, with expectations of further rises to 6% (ONS, 2021). This causes consumption to fall, ceterus paribus, as these people no longer have the disposable income needed to maintain their previous levels of spending.

This effect is exacerbated by a concept termed “animal spirits”, i.e. consumer confidence. The uncertainty about the duration and magnitude of the pandemic, coupled with rising unemployment, damages animal spirits. This causes consumers to increase their precautionary savings – even those with unchanged incomes, who remain pessimistic about their economic outlook. This is supported by a House of Commons report, which found that the savings rate had risen from 7.7% in Q1 2020 to 27.4% in Q2 2020. This “hoarding” necessitates lower levels of spending, so consumption of non-essential goods and services declines (Baldwin, 2020).

Figure 6: COVID-19 effects on essential and non-essential goods markets and GDP



- This graph from Suborna (2020) shows how consumption has fallen and its composition has changed. Demand has risen for essential goods like groceries and medical supplies (A), but has fallen for non-essential goods and services (B). The net effect is a decrease in AD to AD₁, and a decrease in SRAS to SRAS₁ (C).

Many developed countries have used fiscal stimuli to slow falling AD and rising unemployment. The UK government created a furlough scheme to pay 80% of workers' wages, and the U.S. government is delivering a \$1.9tn stimulus in 2021. This alleviated some of the fall in consumer spending, but nevertheless in Q3 2020 it was still down 8.3% on Q4 2019 (UK) (HoC, 2020).

Investment, the sum of purchases which add to the capital stock of firms and households, is another component of AD. Investment is typically sensitive to GDP swings, thus has also fallen during the pandemic. On an autonomous level, shrinking revenue streams coupled with fixed costs means firms simply have smaller cash pools to invest with. And whilst interest rates remain low, low animal spirits mean firms are pessimistic about future demand, therefore view investments as risky. For these reasons, firms invested 10.7% less in 2020 than 2019, and expect to invest 10% less even in Q3 2021 vs a non-Covid scenario. In sum, the supply of loanable funds for investment is up, but demand for these funds is down. As Mohaddes et al. (2020) explain, this is because global economic volatility has exceeded the threshold level which creates these effects, with a significant relationship in 80% of advanced economies.

Long term effects: aggregate supply

Physical capital and growth

The Solow-Swan model is a production function linking output to inputs of capital and labour. The model highlights the importance of capital to output, with capital deepening and improvements in total factor productivity via exogenous technological shocks being the two causes of economic growth (Whelan, 2020). Increases in productivity shift the production function outwards, allowing more output per capita holding capital per capita constant.

In Solow's model, the change in the level of capital per capita is a function of the savings rate (+), as savings equal investment, and the depreciation rate (-). Algebraically:

$$\frac{K_{t+1}}{N} - \frac{K_t}{N} = s \frac{Y_t}{N} - \delta \frac{K_t}{N}$$

Ludvigson et al. (2020) point out that unlike previous natural disasters, Covid-19 has not destroyed physical capital. However, Johnson and Behravesch (2020) explain how reduced investment, coupled with depreciation and rising bankruptcies, might still lower the capital stock. Algebraically:

$$\frac{K_{t+1}}{N} < \frac{K_t}{N}$$

In the Solow-Swan model, savings are equal to investments. However, this essay has already explained how low animal spirits have caused UK investment to fall, despite the ballooning savings rate. So this assumption may not currently be true; however if these animal spirits begin to rise, and the savings rate remains higher due to a cultural shift, we may see the capital stock begin to grow. This would increase the level of output per capita, but would not affect long-term growth rates.

Supporting Johnson and Behravesch, radically altered consumption trends have rendered some specialist capital void (Haldane, 2020). For example, aeroplanes cannot be effectively repurposed to create output whilst airports are forced to close. How quickly repurposing can be achieved will determine economic scarring. McKibbin (2020) sheds light on this, explaining how rigidities in the economy will delay adjustment, such as the cost of adjusting investment, and uncertainty about whether the capital is void or not. Governments must signal this to firms wherever possible, though due to the nature of the virus this is difficult to achieve.

Ultimately, the pandemic may change the level of capital per capita and the long-term savings rate, thereby affecting output per capita. In the long run, however, Solow-Swan posits that growth can only be determined by productivity shocks, and offers limited insight into how these occur.

Nevertheless, Fornaro and Wolf (2020) relate the lower investment rate observed to lower productivity thus output via a 'supply-demand doom loop'. Here, lower demand induces firms to cut investment, which endogenously drops productivity growth as workers have less high-quality capital to work with. This reduces incomes (as wages are determined by the marginal productivity of labour) thus further reducing demand and then investment.

Human capital and growth

A more comprehensive understanding of the pandemic's effect on productivity, thus growth rates, can be obtained by exploring how it might affect the stock of human capital. Human capital might be degraded due to the widespread disruption of education systems. Further, shifting consumption trends might necessitate a significant reallocation of resources, potentially creating structural unemployment.

In the same way that structural shifts can render specialist physical capital void, they may render human capital void by creating skill mismatches. For example, in the aftermath of Covid-19 we might expect resources to be reallocated away from the hospitality industry, and towards IT and e-commerce. Where unemployment becomes structural, workers might drop out of the workforce, experiencing skills degradation and at worst, hysteresis. This could cause long-term economic "scarring", i.e. a reduction in economic potential (Haldane, 2020). The threat of this is greatest in service-orientated economies, especially those reliant on exporting tourism. According to Fernandes (2020), for these countries long-term scarring may be underestimated due to overreliance on historical comparisons with SARS. Conversely, this factor may be of less concern in nations such as Taiwan, which are net exporters in sectors that have grown the pandemic. Taiwan's exports accounted for 60% of its 3% GDP growth in 2020 (Bloomberg, 2021), as its technology sector has benefited from soaring demand during lockdowns.

The pandemic's disruption of the education system could further contribute to declining productivity. In a study focused on the U.S. system, Garcia and Weiss (2020) conclude that online learning is only effective where students have consistent access to the requisite technology, and teachers have received targeted training. As such, it seems intuitive to assume that at least some students, particularly those from lower socioeconomic backgrounds, will have their learning and development impeded due to this disruption (ibid). Applying human capital theory (Becker, 1964), this could inhibit children developing the skills which make them more productive, thus reducing the rate of long-run economic growth, as well as incomes.

Social capital and growth

Flowing from this, the pandemic might damage long-term growth through its effect on social capital, which refers to agents' social networks which share knowledge and generate trust.

Social capital is related to growth via its positive relationship to entrepreneurship, because of its role in diffusing information regarding business opportunities (Kenney, 1988). This relationship is strengthened by microeconomic efficiencies whereby high levels of social capital reduce transaction costs (Putnam, 2001). For these reasons, Beugelsdijk (2004) finds a positive relationship social capital and economic growth in 54 European regions. Further, Aldrich (2012) surmises how communities with high social capital respond more effectively to crises, therefore recover in a more comprehensive and sustainable fashion over time.

Covid-19 could degrade social capital by limiting network interactions due to lockdowns and social distancing. This prevents the casual interactions with heterogeneous individuals which take place in schools, work, and public spaces (Pitas, 2020). Further, if in the long run there are cultural shifts towards remote working, this could disperse interactions away from dense urban areas (The Economist, 2020) and encourage people to spend more time indoors. This might limit the exchange of knowledge and ideas that contributes to entrepreneurship and economic growth. Pitas (2020) takes this further, explaining how an increasing reliance on digital communications reinforces existing

relationships, called bonding social capital, as opposed to meaningful 'bridging' interactions with heterogeneous individuals.

Linking to this, the pandemic has caused a de-globalisation process, with lockdowns restricting transnational flows of capital and labour. If anti-globalisation sentiment rises due to the pandemic, this could extend into a fresh wave of populism. This would harm long-term economic growth through two mechanisms: the loss of comparative advantage (Ricardo, 1817) and reduced knowledge diffusion per the Lucas model of growth, due to excessive bonding relative to bridging social capital. Granovetter (1973) points out that this can inhibit innovation, thus growth, because of clan mentalities where groups resent new information.

Determining factors

The pandemic will affect growth by impacting each form of capital and thus productivity. However, there are a range of factors which might alter this relationship.

Vaccine roll-out

One is the effectiveness of vaccination programmes, namely the speed and scope of rollouts. First, a strong vaccine programme should raise animal spirits, in turn unleashing the build-up of savings in developed economies. The UK's higher savings rate will result in an extra £250bn of private savings by June 2021 (House of Commons, 2020); inducing consumers to convert this into spending can create a demand-led recovery. This improved consumer sentiment should also induce investment by reducing uncertainty (Haldane, 2020), further boosting aggregate demand and supply, and limiting medium term economic damage.

Secondly, an effective vaccine programme gives policymakers more scope to prevent scarring. Drawing upon previous analyses, the quicker in-person education and training can be resumed, the less human capital will depreciate. The quicker governments can wind down fiscal stimuli without increasing unemployment, the less public debt is accumulated.

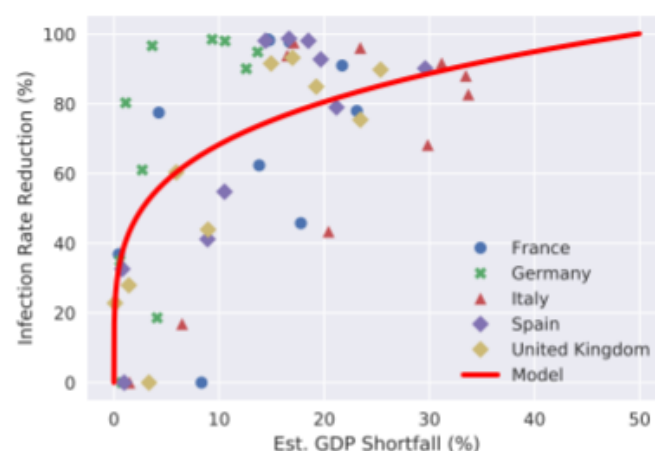
It is important to qualify these points by highlighting the importance of global vaccine programmes, as well as domestic ones. These can determine trade and investment flows by improving sentiments and incomes in other countries (Suborna, 2020). These points are underscored by the Office for Budget Responsibility (OBR), which outlined three scenarios based on the effectiveness of the vaccine programme. With an effective programme, GDP returns to pre-virus levels by Q4 2021, with no long-term damage to the economy. In the absence of an effective vaccine, the OBR predicts lasting long-term damage to the economy of 6% of GDP.

Policy effectiveness

Many governments have been borrowing in record amounts to stimulate their economies, including the UK in 2020 (ONS, 2020). This, coupled with falling GDP, is increasing the debt-to-GDP ratio of most countries. This is concerning, as Mohaddes et al (2018) find rising debt-to-GDP ratios are associated with lower long-term growth, due to the crowding out of private sector investment, and falling investor confidence leading to higher long-term borrowing rates. Elevated public debt will also reduce the ability of governments to pursue expansionary fiscal policies in future downturns (Kral, 2021). These concerns are even more valid considering the majority of UK government spending has been current spending i.e., the furlough scheme, which is not growth-enhancing like capital spending is.

For these reasons, governments may have to use austerity post-Covid, but this comes with its own problems. Kral (2021) points out that austerity typically weighs down growth rates, for example due to higher taxes disincentivising investment, or reduces welfare, for example due to government spending cuts. Therefore, policies must strive to support short and long-term growth without accumulating unnecessary debt.

Linking to this, Trotter (2020) highlights the need for policies to balance tackling the virus with sustaining GDP levels. His model, visualised below, shows that infection rate reductions exhibit decreasing returns in the GDP shortfall, because policies which are most effective at reducing infections and least disruptive to the economy are used first. Understanding where we lie on this curve could be a useful step toward understanding which policies might be proportional in terms of economic scarring incurred.



Potential for green recoveries

An indirect effect of Covid-19 on long-term growth is the impetus it might generate for a “green recovery” which lays the foundations for sustainable growth that curtails climate change. The importance of tackling climate change is highlighted by Mohaddes et al. (2019), who find that labour productivity is affected by consistent temperature deviations from historical norms. Further, they find climate change can increase the rate of capital depreciation by increasing the frequency of natural disasters. Applying this to a growth model in which output growth is positively affected by technological progress and negatively affected by deviations of climate variables, they find that a temperature change of 0.04 degrees per annum would reduce real GDP per capita by 7% by 2100.

This relates to Covid-19 as during the first lockdowns in Q1-Q2 2020, global greenhouse gas (GHG) emissions fell by 13.6% (The Guardian, 2020), which increased environmental awareness. Maillet et al

(2020) propose building upon this momentum with a lump sum carbon tax to create a so-called double dividend. By changing relative prices of carbon-intensive energy and renewable energy, a carbon tax incentivises the substitution of fossil fuels for capital, reducing GHGs and stimulating energy efficiency investments (ibid). Secondly, the tax revenues can be recycled as a lump sum tax credit for sustainable investments, which boost short-term employment, generate economies of scale, and increase productivity (Stern, 2006). This macroeconomic benefit is even greater for net oil importing countries because raising oil prices will shift consumption from imported oil to domestically produced renewables, improving the country's balance of payments. In the EU, on average 87% of oil consumption is imported (Eurostat, 2019).

So, will long run growth suffer?

Most countries have experienced lower short-term growth rates during the pandemic due to contractions in aggregate demand and short-run aggregate supply, which reduce real GDP and have an ambiguous effect on price levels.

The long run economic effects of Covid-19 are less certain: a range of variables that might accentuate or attenuate the way in which the pandemic affects capital stocks and productivity. These include vaccine programmes and government policies, in terms of the future budget constraints they create, and whether they channel resources into a green recovery to avoid the impacts of climate change on long-run growth.

That said, most commentators believe the aggregate long-term impact will be negative, with Ramsden (2020) predicting a reduction in the supply capacity of the UK economy of 1.75%. This will reduce the rate of growth at which our economy can grow without accelerating inflation. Further, Mohaddes et al. (2020) hypothesise that the UK will experience deeper and longer-lasting effects than the global average, which they believe will see global real GDP 3% than in a no pandemic scenario.

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